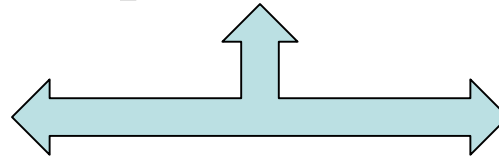


9.4 the solution of linear time-invariant system state equation

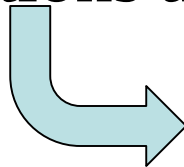
Mathematical
Representation

Quantitative(定量)
Analysis

Qualitative(定性)
Analysis



Solving the dynamic mathematical model equations;
The solutions analysis.



Transfer function

State Space – modern control theory



Matrices Operation (State transition Matrix)
Dynamic Response

State equation (Model) $\implies$ Dynamic analysis (solve state equation)
Insuring the existence and uniqueness of the solution: the elements in A and B are bounded.

9.4.1 Solution of Linear Time-invariant Continual System

1. The solution of **homogeneous state equation**(齐次状态方程):

$\dot{x} = Ax$ is homogeneous state equation, and there are general 2 solutions:

➤ **Power Series Method** (幂级数法)

Assume the solution of above **equation** is a vector power series (幂级数) of t

$$x(t) = b_0 + b_1 t + b_2 t^2 + \cdots + b_k t^k + \cdots$$

$x, b_0, b_1, \cdots, b_k \cdots$ are n dimensional vectors.

Calculate the derivative of above equation:

$$\dot{x} = b_1 + 2b_2t + \cdots + kb_kt^{k-1} + \cdots = A(b_0 + b_1t + b_2t^2 + \cdots + b_kt^k + \cdots)$$

Assume the coefficients with the same power are uniform.

$$b_1 = Ab_0$$

$$b_2 = \frac{1}{2} Ab_1 = \frac{1}{2} A^2 b_0$$

$$b_3 = \frac{1}{3} Ab_2 = \frac{1}{3 \times 2} A^3 b_0$$

$$\vdots$$

$$b_k = \frac{1}{k} Ab_{k-1} = \frac{1}{k!} A^k b_0$$

$$\vdots$$

$$\therefore x(0) = b_0$$

$$\therefore x(t) = (I + At + \frac{1}{2} A^2 t^2 + \cdots + \frac{1}{k!} A^k t^k + \cdots) x(0)$$

Define:

$$e^{At} = I + At + \frac{1}{2} A^2 t^2 + \cdots + \frac{1}{k!} A^k t^k + \cdots = \sum_{k=0}^{\infty} \frac{1}{k!} A^k t^k$$

$$x(t) = e^{At} x(0)$$

e^{At} ——— Matrix exponential function, called state transition matrix: $\Phi(t)$.₃

➤ Laplace transformation for $\dot{x} = Ax$

$$sx(s) - x(0) = Ax(s)$$

$$(Is - A)x(s) = x(0)$$

$$x(s) = (Is - A)^{-1}x(0)$$

Inverse Laplace Transformation:

$$x(t) = L^{-1}[(sI - A)^{-1}]x(0)$$

Compare with the power series method:

$$\Phi(t) = e^{At} = L^{-1}[(sI - A)^{-1}]$$

↑

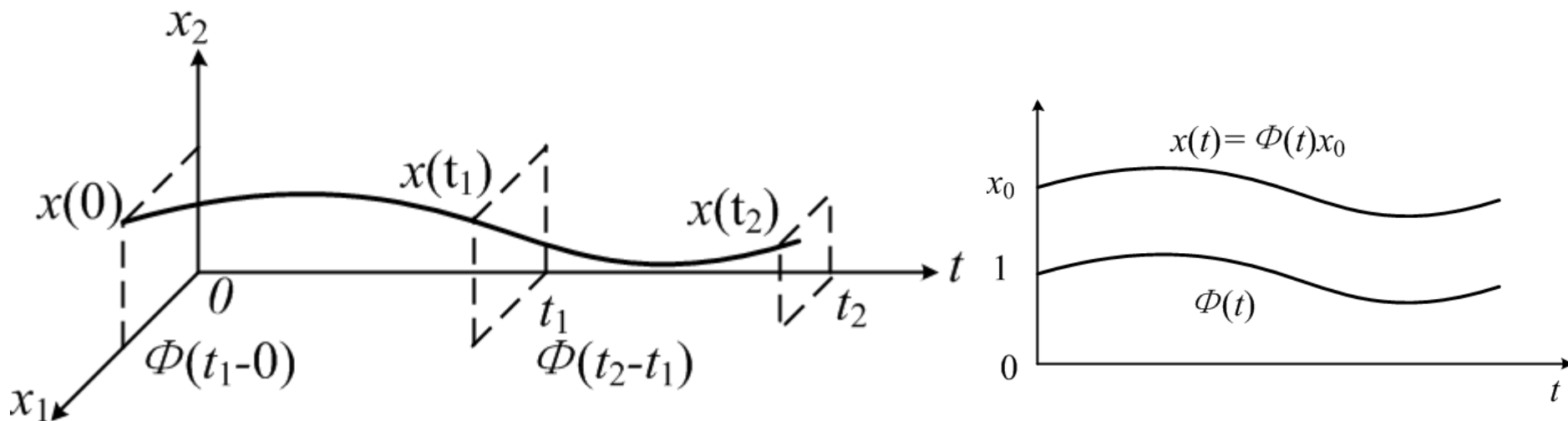
The closed form (闭合形式) (analytic form 解析形式) of the state transition matrix, which is convergent (收敛的).

solve homogeneous state equation $\implies$ calculate state transition matrix

Discussion:

$$\dot{x} = Ax \Rightarrow x(t) = e^{At} x(0) \text{ OR } x(t) = e^{A(t-t_0)} x(t_0)$$

- The solution of homogeneous state equation describe a freedom motion (自由运动) of the system without the input $u(t)$, which is the transition of the initial state only based on the state transition matrix $e^{A(t-t_0)}$.



2. The solution of **non-homogeneous state equation**:

Give the non-homogeneous state equation:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad \mathbf{x}(t) \in R^n, \mathbf{u}(t) \in R^r, \mathbf{A} \in R^{n \times n}, \mathbf{B} \in R^{n \times r}$$

(1) Direct method (Integral method 积分法) $\dot{\mathbf{x}}(t) - \mathbf{A}\mathbf{x}(t) = \mathbf{B}\mathbf{u}(t)$

left multiply $e^{-\mathbf{A}t}$ simultaneously: $\underline{e^{-\mathbf{A}t}[\dot{\mathbf{x}}(t) - \mathbf{A}\mathbf{x}(t)] = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t)}$

$$\frac{d}{dt}[e^{-\mathbf{A}t}\mathbf{x}(t)] = e^{-\mathbf{A}t}\mathbf{B}\mathbf{u}(t)$$

$$\boxed{\frac{d}{dt}[e^{-\mathbf{A}t}\mathbf{x}(t)]}$$

$$e^{-\mathbf{A}t}\mathbf{x}(t) - \mathbf{x}(0) = \int_0^t e^{-\mathbf{A}\tau}\mathbf{B}\mathbf{u}(\tau)d\tau$$

$$\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \quad \mathbf{x}(t)|_{t=0} = \mathbf{x}(0)$$

$$\mathbf{x}(t) = \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t-\tau)\mathbf{B}\mathbf{u}(\tau)d\tau$$

response of initial condition

Response of input $\mathbf{u}(t)$

(2) Laplace transformation method

$$sX(s) - x(0) = AX(s) + Bu(s)$$

$$(sI - A)X(s) = x(0) + Bu(s)$$

$$X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}Bu(s)$$

then $x(t) = L^{-1}[(sI - A)^{-1}x(0)] + L^{-1}[(sI - A)^{-1}Bu(s)]$

(from $e^{At} = L^{-1}[(sI - A)^{-1}]$, we have)

$$x(t) = e^{At}x(0) + \int_0^t e^{A(t-\tau)}Bu(\tau)d\tau = \Phi(t)x(0) + \int_0^t \Phi(t - \tau)Bu(\tau)d\tau$$

Discussion:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \implies \mathbf{x}(t - t_0) = \Phi(t - t_0)\mathbf{x}(t_0) + \int_{t_0}^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau$$

- The solution of **non-homogeneous state equation** is composed by two parts
 - the freedom motion of the initial state: $\Phi(t-t_0)\mathbf{x}(t_0)$, which is called zero-input response;
 - the controlled motion by the input: $\int_0^t \Phi(t-\tau)\mathbf{B}\mathbf{u}(\tau)d\tau$, which is called zero-state response.

9.4.2 Properties of state transition matrix

$$e^{At} = I + At + \frac{1}{2}A^2t^2 + \cdots + \frac{1}{k!}A^kt^k + \cdots = \sum_{k=0}^{\infty} \frac{1}{k!}A^kt^k$$

1. Initial state: $\Phi(0) = I$
2. $\dot{\Phi}(t) = A\Phi(t) = \Phi(t)A$ $\dot{\Phi}(0) = A$
3. Linear relationship: $\Phi(t_1 \pm t_2) = \Phi(t_1)\Phi(\pm t_2) = \Phi(\pm t_2)\Phi(t_1)$
4. Reversibility: $\Phi^{-1}(t) = \Phi(-t)$, $\Phi^{-1}(-t) = \Phi(t)$
5. $x(t_2) = \Phi(t_2 - t_1)x(t_1)$
6. Segmentation: $\Phi(t_2 - t_0) = \Phi(t_2 - t_1)\Phi(t_1 - t_0)$
7. $[\Phi(t)]^k = \Phi(kt)$
8. if $AB = BA$, $e^{(A+B)t} = e^{At}e^{Bt} = e^{Bt}e^{At}$;
if $AB \neq BA$, $e^{(A+B)t} \neq e^{At}e^{Bt} \neq e^{Bt}e^{At}$

9. if $\Phi(t)$ is state transition matrix of $\dot{x}(t) = Ax(t)$, the newly state transition matrix after non-singular transform $x = P\bar{x}$ is:

$$\bar{\Phi}(t) = P^{-1}e^{At}P$$

10. Two common state transition matrices

If A is n -order Diagonal Matrix,

$$A = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix}, \quad \Phi(t) = \begin{bmatrix} e^{\lambda_1 t} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & e^{\lambda_n t} \end{bmatrix}$$

If A is m -order Jordan Matrix,

$$A = \begin{bmatrix} \lambda & 1 & \cdots & 0 \\ 0 & \lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 0 & \cdots & 0 & \lambda \end{bmatrix}_{m \times m}, \quad \Phi(t) = \begin{bmatrix} e^{\lambda t} & te^{\lambda t} & \cdots & \frac{t^{m-1}}{(m-1)!}e^{\lambda t} \\ 0 & e^{\lambda t} & \ddots & \vdots \\ \vdots & \ddots & \ddots & te^{\lambda t} \\ 0 & \cdots & 0 & e^{\lambda t} \end{bmatrix}$$

9.4.3 Calculation of matrix transition function e^{At}

- **Method One: Direct method** (matrix power function)

$$e^{At} = I + At + \frac{A^2 t^2}{2!} + \frac{A^3 t^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{1}{k!} A^k t^k$$

For any constant matrix A and limited t , the above infinite series is convergent.

- **Method Two: Linear transform method** (diagonal form method and Jordan form method)

If the matrix A can be transited to the diagonal form, e^{At} can be given as:

$$e^{At} = P e^{\Lambda t} P^{-1} = P \begin{bmatrix} e^{\lambda_1 t} & & & 0 \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ 0 & & & e^{\lambda_n t} \end{bmatrix} P^{-1}$$

In the above equation, P is the non-singular linear transform matrix for A .

Similarly, if matrix A can be transformed to Jordan form, e^{At} can be given as:

$$e^{At} = Se^{Jt}S^{-1}$$

➤ Method Three: Laplace transform method

$$e^{At} = L^{-1}[(sI - A)^{-1}]$$

For e^{At} , it is essential to calculate the inverse of $(sI-A)$. Generally, the Recursive Algorithm (递推算法) can be used when the order of system matrix A is high.

Ex.9-13 Consider following system matrix, try to find the proper e^{At} by linear transform method and Laplace transform method.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix}$$

Solution:

Linear transform method: the eigenvalues of A is 0 and -2 ($\lambda_1=0$, $\lambda_2=-2$), thus, the transform matrix P is:

$$P = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix}$$

from
$$e^{At} = P e^{\Lambda t} P^{-1} = P \begin{bmatrix} e^{\lambda_1 t} & & & 0 \\ & e^{\lambda_2 t} & & \\ & & \ddots & \\ 0 & & & e^{\lambda_n t} \end{bmatrix} P^{-1}$$

we have,
$$e^{At} = \begin{bmatrix} 1 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} e^0 & 0 \\ 0 & e^{-2t} \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2}(1 - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix}$$

Laplace transform method:

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 0 & s + 2 \end{bmatrix}$$

We have:

$$(sI - A)^{-1} = \begin{bmatrix} \frac{1}{s} & \frac{1}{s(s+2)} \\ 0 & \frac{1}{s+2} \end{bmatrix}$$

thus:

$$e^{At} = L^{-1}[(sI - A)^{-1}] = \begin{bmatrix} 1 & \frac{1}{2}(1 - e^{-2t}) \\ 0 & e^{-2t} \end{bmatrix}$$

Ex.9-14 find the state transition matrix $\Phi(t)$ and its inverse of following linear time-invariant system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Solution:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$\Phi(t) = e^{At} = L^{-1}[(sI - A)^{-1}]$$

$$sI - A = \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 \\ 2 & s+3 \end{bmatrix}$$

$$(sI - A)^{-1} = \frac{1}{(s+1)(s+2)} \begin{bmatrix} s+3 & 1 \\ -2 & s \end{bmatrix} = \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix}$$

$$\begin{aligned}\Phi(t) &= e^{At} = L^{-1}[(sI - A)^{-1}] \\ &= \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}\end{aligned}$$

Then calculate the inverse of state transition matrix $\Phi^{-1}(t)$.

According to $\Phi^{-1}(t) = \Phi(-t)$, the inverse of state transition matrix is:

$$\Phi^{-1}(t) = e^{-At} = \begin{bmatrix} 2e^t - e^{2t} & e^t - e^{2t} \\ -2e^t + 2e^{2t} & -e^t + 2e^{2t} \end{bmatrix}$$

Ex.9-15 try to find the time response relationship of following system, in which, the input $u(t)=1(t)$, the unit step function at $t=0$.

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Solution:

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\Phi(t) = e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \quad (\text{according to [Ex.9-14](#)})$$

$$x(t) = e^{At} x(0) + \int_0^t \begin{bmatrix} 2e^{-(t-\tau)} - e^{-2(t-\tau)} & e^{-(t-\tau)} - e^{-2(t-\tau)} \\ -2e^{-(t-\tau)} + 2e^{-2(t-\tau)} & -e^{-(t-\tau)} + 2e^{-2(t-\tau)} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} 1(t) d\tau$$

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

If the initial state is zero: $\mathbf{x}(0)=\mathbf{0}$, $\mathbf{x}(t)$ can be simplified as:

$$\begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix} = \begin{bmatrix} \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t} \\ e^{-t} - e^{-2t} \end{bmatrix}$$

Ex.9-16 Assume the dynamic equation is: $\ddot{y} + (a+b)\dot{y} + aby = \dot{u} + cu$

With a, b and c are real constants. Try to find:

- (1) the state space equation of the system;
- (2) the state transition matrix $\Phi(t)$.

Solution:

$$\begin{aligned} (1) \quad G(s) &= \frac{Y(s)}{U(s)} = \frac{s+c}{s^2 + (a+b)s + ab} \\ &= \frac{s+c}{(s+a)(s+b)} \\ &= \frac{c-a}{b-a} \cdot \frac{1}{s+a} + \frac{c-b}{a-b} \cdot \frac{1}{s+b} \end{aligned} \quad \begin{cases} \dot{x} = \begin{bmatrix} -a & 0 \\ 0 & -b \end{bmatrix} x + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u \\ y = \begin{bmatrix} \frac{c-a}{b-a} & \frac{c-b}{a-b} \end{bmatrix} x \end{cases}$$

$$(2) \quad \Phi(t) = L^{-1}[(sI - A)^{-1}]$$

$$= L^{-1} \left[\begin{pmatrix} s+a & 0 \\ 0 & s+b \end{pmatrix}^{-1} \right] = L^{-1} \begin{bmatrix} \frac{1}{s+a} & 0 \\ 0 & \frac{1}{s+b} \end{bmatrix} = \begin{bmatrix} e^{-at} & 0 \\ 0 & e^{-bt} \end{bmatrix}$$

9.4.4 Establishing and solution of linear discrete system state space representation

➤ 1. The state space description of discrete-time linear system:

The discrete-time linear time-variant system:

$$x(k+1) = A(k)x(k) + B(k)u(k)$$

$$y(k) = C(k)x(k) + D(k)u(k)$$

The discrete-time linear time-invariant system:

$$x(k+1) = Ax(k) + Bu(k)$$

$$y(k) = Cx(k) + Du(k)$$

$A_{n \times n}$: system matrix;	$B_{n \times p}$: input matrix
$C_{q \times n}$: output matrix;	$D_{q \times p}$: transfer matrix

Population distribution Issue:

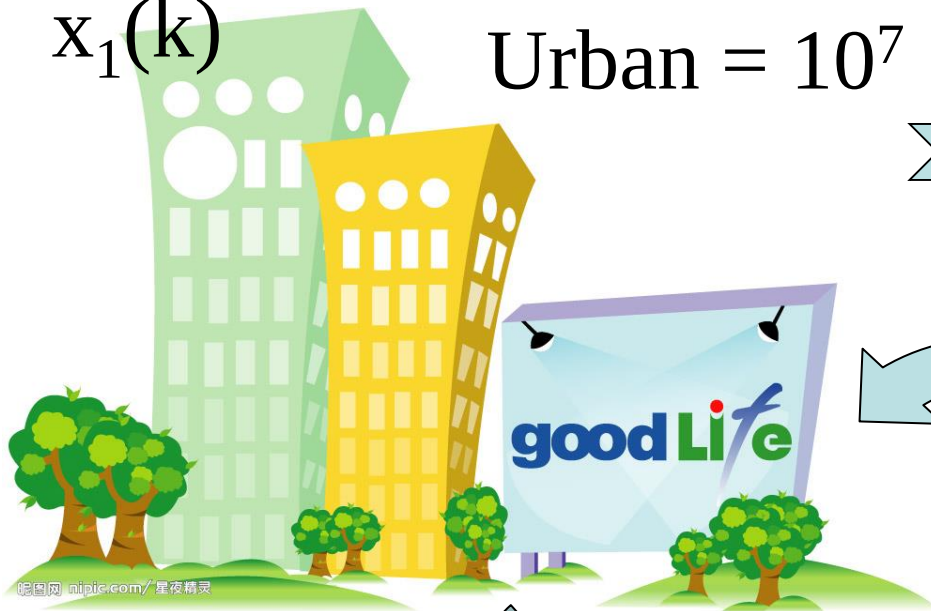
Assume the population condition of a country: Urban population is 10^7 ; Rural population is 9×10^7 ; 4% of urban population transfer to countryside; 2% of rural population transfer to city; increasing rate of the whole country is 1%.

Assume: (1) k is a discrete-time variable; (2) $x_1(k)$ and $x_2(k)$ are urban and rural population of the k^{th} year; (3) $u(k)$ is population control device of the government: a unit of positive control device can inspire 5×10^4 population move from city to the countryside, and v.v. ; (4) $y(k)$ is the total population of the k^{th} year.

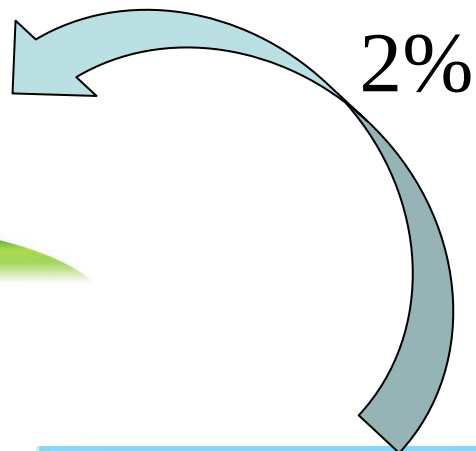
Try to describe such population distribution issue by discrete state space form.

$x_1(k)$

Urban = 10^7



$y(k)$ 1%

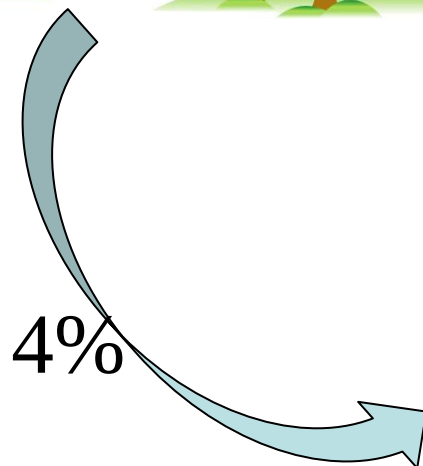


2%

$u(t)$



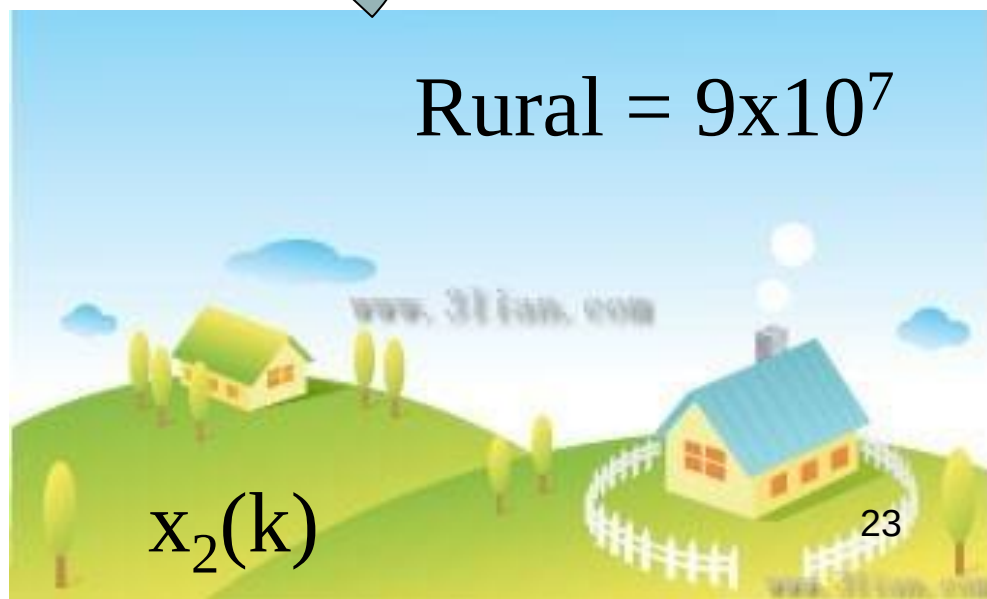
4%



5×10^4



Rural = 9×10^7



$x_2(k)$

Discrete state equations

$$x_1(k+1) = (1+0.01) \times \left[(1-0.04)x_1(k) + 0.02x_2(k) - 5 \times 10^4 u(k) \right]$$

$$x_2(k+1) = (1+0.01) \times \left[(1-0.02)x_2(k) + 0.04x_1(k) + 5 \times 10^4 u(k) \right]$$

Matrix representation:

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \end{bmatrix} = \begin{bmatrix} 0.9696 & 0.0202 \\ 0.0404 & 0.9898 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix} + \begin{bmatrix} -5.05 \times 10^4 \\ 5.05 \times 10^4 \end{bmatrix} u(k)$$

Also standard state space description:

$$y(k) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \end{bmatrix}$$

$$x(k+1) = Ax(k) + Bu(k)$$

$$y(k) = Cx(k) + Du(k)$$

➤ 2. Discrete-time state space from differential equations

The difference equation and impulse transfer function are widely used to describe the discrete system in classic control theory. The general form of time-invariant differential equation of SISO system is:

$$\begin{aligned} y(k+n) + a_1 y(k+n-1) + \cdots + a_{n-1} y(k+1) + a_n y(k) \\ = b_0 u(k+n) + b_1 u(k+n-1) + \cdots + b_{n-1} u(k+1) + b_n u(k) \end{aligned}$$

In which, k is time of kT ; T is sampling period; $u(k)$ and $y(k)$ are input and output at time of kT ; a_i and b_i are constant to describe system performance; consider the Z-transform with zero initial condition:

$$Z[y(k)] = y(z), \quad Z[(y(k+i))] = z^i y(z)$$

$$G(z) = \frac{y(z)}{u(z)} = \frac{b_0 z^n + b_1 z^{n-1} + \cdots + b_{n-1} z + b_n}{z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n}$$

$$= b_0 + \frac{\beta_1 z^{n-1} + \beta_2 z^{n-2} + \cdots + \beta_{n-1} z + \beta_n}{z^n + a_1 z^{n-1} + \cdots + a_{n-1} z + a_n} = b_0 + \frac{N(z)}{D(z)}$$

Such $G(z)$ is **impulse transfer function**, which is similar with the form of continual system:

$$Y(s) = (b_0 s^n + b_1 s^{n-1} + \cdots + b_{n-1} s + b_n) E(s)$$

$$U(s) = (s^n + a_1 s^{n-1} + \cdots + a_{n-1} s + a_n) E(s)$$

The same way in continual system can be used in discrete situation:

Such as intermediate variable method:

Import intermediate variable $Q(z)$ in $N(z)/D(z)$ serial decomposition:

$$u(z) = z^n Q(z) + a_1 z^{n-1} Q(z) + \cdots + a_{n-1} z Q(z) + a_n Q(z)$$

$$y(z) = \beta_1 z^{n-1} Q(z) + \cdots + \beta_{n-1} z Q(z) + \beta_n Q(z)$$

Define state variables:

$$\begin{cases} x_1(z) = Q(z) \\ x_2(z) = zQ(z) = zx_1(z) \\ \vdots \\ x_n(z) = z^{n-1}Q(z) = zx_{n-1}(z) \end{cases}$$

$$z^n Q(z) = -a_n x_1(z) - a_{n-1} x_2(z) \cdots - a_1 x_n(z) + u(z)$$

$$y(z) = \beta_n x_1(z) + \beta_{n-1} x_2(z) + \cdots + \beta_1 x_n(z)$$

Z inverse transform $\quad Z^{-1}[x_i(z)] = x_i(k), \quad Z^{-1}[zx_i(z)] = x_i(k+1)$

$$\begin{cases} x_1(k+1) = x_2(k) \\ x_2(k+1) = x_3(k) \\ \vdots \\ x_{n-1}(k+1) = x_n(k) \\ x_n(k+1) = -a_n x_1(k) - a_{n-1} x_2(k) \cdots - a_1 x_n(k) + u(k) \end{cases}$$

$$y(k) = \beta_n x_1(k) + \beta_{n-1} x_2(k) + \cdots + \beta_1 x_n(k)$$

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ \vdots \\ x_{n-1}(k+1) \\ x_n(k+1) \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_{n-1}(k) \\ x_n(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [\beta_n \quad \beta_{n-1} \quad \cdots \quad \beta_1] \begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_{n-1}(k) \\ x_n(k) \end{bmatrix} + Du(k)$$

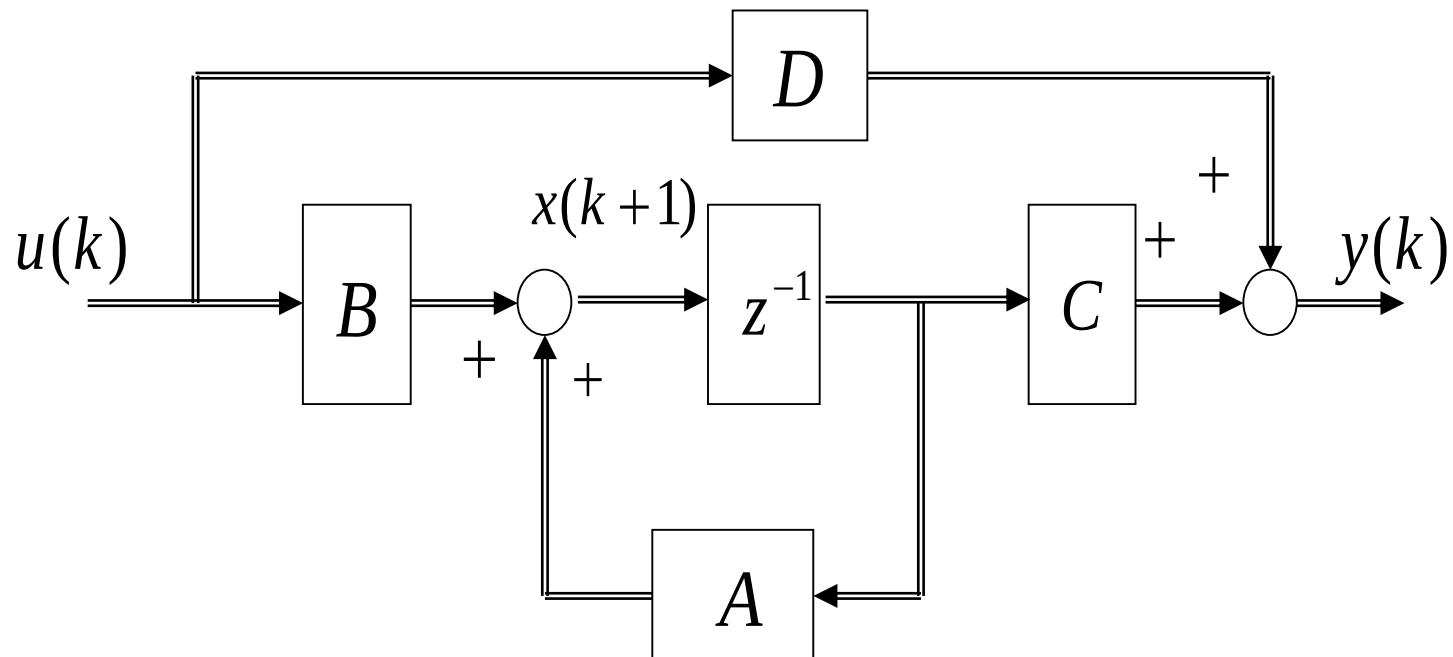
Discrete system state equation describes the relationship between the state of the system at $(k+1)T$, and the state at time kT and input of the system.

Output equation describes the relationship between the output of the system at kT , and the state at kT and input of the system.

State space representation of linear time-invariant MIMO discrete system

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k)$$

$$\mathbf{y}(k) = \mathbf{C}\mathbf{x}(k) + \mathbf{D}\mathbf{u}(k)$$



➤ 3. Discretization of continual system state space expression

The solution of the time-invariant continual state equation $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$ under the input $\mathbf{u}(t)$

$$\mathbf{x}(t) = e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B}\mathbf{u}(\tau) d\tau = \Phi(t-t_0) \mathbf{x}(t_0) + \int_{t_0}^t \Phi(t-\tau) \mathbf{B}\mathbf{u}(\tau) d\tau$$

assume $t_0 = kT$, $\mathbf{x}(t_0) = \mathbf{x}(kT) = \mathbf{x}(k)$

$t = (k+1)T$, $\mathbf{x}(t) = \mathbf{x}[(k+1)T] = \mathbf{x}(k+1)$

at $t \in [k, k+1]$, $\mathbf{u}(k) = \mathbf{u}(k+1)$ is constant

$$\mathbf{x}(k+1) = \Phi[(k+1)T - kT] \mathbf{x}(k) + \left(\int_{kT}^{(k+1)T} \Phi[(k+1)T - \tau] \mathbf{B} d\tau \right) \mathbf{u}(k)$$

$$\mathbf{G}(T) = \int_{kT}^{(k+1)T} \Phi[(k+1)T - \tau] \mathbf{B} d\tau$$

Variable replacement $(k+1)T - \tau = \tau'$

then

$$\mathbf{G}(T) = \int_0^T \Phi(\tau) \mathbf{B} d\tau$$

$$\mathbf{x}(k+1) = \Phi(T)\mathbf{x}(k) + \left(\int_0^T \Phi(\tau)B d\tau \right) \mathbf{u}(k)$$

State equation of discrete system is:

$$\mathbf{x}(k+1) = \Phi(T)\mathbf{x}(k) + G(T)\mathbf{u}(k)$$

The relationship between $\Phi(T)$ and state transition matrix $\Phi(t)$ of continual system:

$$\Phi(T) = \Phi(t) \Big|_{t=T}$$

The output equation of discrete system is:

$$y(k) = C\mathbf{x}(k) + D\mathbf{u}(k)$$

Ex.9-17 find the discrete state equation with $T=1s$ from following continual system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Solution: from [Ex.9-15](#), the state transition matrix $\Phi(t)$ of above continual system is:

$$\Phi(t) = e^{At} = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

$$\Phi(T) = \Phi(t)|_{t=T=1} = \begin{bmatrix} 0.6004 & 0.2325 \\ -0.4651 & -0.0972 \end{bmatrix}$$

$$G(T) = \int_0^T \Phi(\tau) B d\tau = \int_0^T \begin{pmatrix} e^{-\tau} - e^{-2\tau} \\ -e^{-\tau} + 2e^{-2\tau} \end{pmatrix} d\tau = \begin{bmatrix} 1/2 - e^{-T} + 1/2e^{-2T} \\ e^{-T} - e^{-2T} \end{bmatrix}$$

$$G(T)|_{T=1} = \begin{bmatrix} 0.1998 \\ 0.2325 \end{bmatrix}$$

$$\mathbf{x}(k+1) = \Phi(T)\mathbf{x}(k) + G(T)\mathbf{u}(k)$$

➤4. Solution of time-invariant discrete system dynamic equation

(1) Recurrence method (递推法)

(2) Z transform method

$$\mathbf{x}(k+1) = \Phi(T)\mathbf{x}(k) + G(T)\mathbf{u}(k) \quad k = 0, 1, \dots, k-1,$$

The states at time of $T, 2T, \dots, kT$ time:

$$k=0 \quad \mathbf{x}(1) = \Phi(T)\mathbf{x}(0) + G(T)\mathbf{u}(0)$$

$$\begin{aligned} k=1 \quad \mathbf{x}(2) &= \Phi(T)\mathbf{x}(1) + G(T)\mathbf{u}(1) \\ &= \Phi^2(T)\mathbf{x}(0) + \Phi(T)G(T)\mathbf{u}(0) + G(T)\mathbf{u}(1) \end{aligned}$$

$$\begin{aligned} k=2 \quad \mathbf{x}(3) &= \Phi(T)\mathbf{x}(2) + G(T)\mathbf{u}(2) \\ &= \Phi^3(T)\mathbf{x}(0) + \Phi^2(T)G(T)\mathbf{u}(0) + \Phi(T)G(T)\mathbf{u}(1) + G(T)\mathbf{u}(2) \end{aligned}$$

$$\vdots \quad \quad \quad \vdots$$

$$\begin{aligned}
 \mathbf{x}(k) &= \Phi(T)\mathbf{x}(k-1) + G(T)\mathbf{u}(k-1) \\
 k = k-1 \quad &= \Phi^k(T)\mathbf{x}(0) + \Phi^{k-1}(T)G(T)\mathbf{u}(0) + \Phi^{k-2}(T)G(T)\mathbf{u}(1) + \cdots \\
 &\quad + \Phi(T)G(T)\mathbf{u}(k-2) + G(T)\mathbf{u}(k-1) \\
 &= \Phi^k(T)\mathbf{x}(0) + \sum_{i=0}^{k-1} \Phi^{k-i-1}(T)G(T)\mathbf{u}(i)
 \end{aligned}$$

It is the solution of discrete state equation, which is **Discrete State Transition Equation**.

when $u(i) = 0, (i = 0, 1, \dots, k-1)$

$$\mathbf{x}(k) = \Phi^k \mathbf{x}(0) = \Phi(kT) \mathbf{x}(0) = \Phi(k) \mathbf{x}(0)$$

$\Phi(k) \implies$ **State transition matrix** of discrete system

The output equation:

$$\begin{aligned} \mathbf{y}(k) &= C\mathbf{x}(k) + D\mathbf{u}(k) \\ &= C\Phi^k(T) \mathbf{x}(0) + C \sum_{i=0}^{k-1} \Phi^{k-i-1}(T) G(T) \mathbf{u}(i) + D\mathbf{u}(k) \end{aligned}$$

For the following discrete state equation:

$$\mathbf{x}(k+1) = \mathbf{A}\mathbf{x}(k) + \mathbf{B}\mathbf{u}(k)$$

$$\mathbf{y}(k) = C\mathbf{x}(k) + D\mathbf{u}(k)$$

Its solution is:

$$\begin{aligned} \mathbf{x}(k) &= A^k \mathbf{x}(0) + \sum_{i=0}^{k-1} A^{k-i-1} B \mathbf{u}(i) \\ \mathbf{y}(k) &= C A^k \mathbf{x}(0) + C \sum_{i=0}^{k-1} A^{k-i-1} B \mathbf{u}(i) + D \mathbf{u}(k) \end{aligned}$$

Ex.9-18 the state equation of the linear discrete system

$$A = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x(k+1) = Ax(k) + Bu(k)$$

Find its solution by using the recurrence method with

the initial state: $x(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

and the input: $u(k) = 1 \quad (k \geq 0)$

Solution:

Recurrence method:

when $k = 0$

$$\begin{aligned} x(1) &= Ax(0) + Bu(0) \\ &= \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \times 1 = \begin{bmatrix} 0 \\ 1.84 \end{bmatrix} \end{aligned}$$

when $k = 1$

$$x(2) = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1.84 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \times 1 = \begin{bmatrix} 2.84 \\ -0.84 \end{bmatrix}$$

when $k = 2$

$$x(3) = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix} \begin{bmatrix} 2.84 \\ -0.84 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \times 1 = \begin{bmatrix} 0.16 \\ 1.39 \end{bmatrix}$$

when $k = 3$

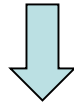
$$x(4) = \begin{bmatrix} 0 & 1 \\ -0.16 & -1 \end{bmatrix} \begin{bmatrix} 0.16 \\ 1.39 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \times 1 = \begin{bmatrix} 2.39 \\ -0.41 \end{bmatrix}$$

Iterate the operation, and we have $x(k)$ at any sampling time.

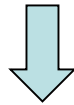
We **cannot** obtain the solutions of the linear discrete equation with **closed-form** by the **recurrence method**, rather than use **state transition matrix**.

- Z transform:

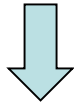
$$x(k+1) = \Phi x(k) + Gu(k)$$



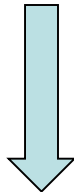
$$zX(z) - zx(0) = \Phi X(z) + GU(z)$$



$$(zI - \Phi)X(z) = zx(0) + GU(z)$$



$$X(z) = (zI - \Phi)^{-1}[zx(0) + GU(z)]$$



$$\begin{aligned} x(k) &= Z^{-1}[X(z)] = Z^{-1}\{(zI - \Phi)^{-1}[zx(0) + GU(z)]\} \\ &= Z^{-1}[(zI - \Phi)^{-1}zx(0)] + Z^{-1}[(zI - \Phi)^{-1}GU(z)] \end{aligned}$$

Analysis:

- $Z^{-1}[(zI - \Phi)^{-1}zx(0)]$

For scale quantity a : $Z^{-1}\left[\frac{1}{1-az^{-1}}\right] = a^k$

Similarly for matrix Φ :

$$Z^{-1}\left[(zI - \Phi)^{-1}z\right] = Z^{-1}\left[(1 - \Phi z^{-1})^{-1}\right] = \Phi^k$$

- $Z^{-1}[(zI - \Phi)^{-1}GU(z)]$

For scale function $w(k)$, and relative function $W(z)$

$$Z^{-1}\{W_1(z)W_2(z)\} = \sum_{i=0}^k w_1(k-i)w_2(i)$$

Thus:

$$Z^{-1}[(zI - \Phi)^{-1}GU(z)] = Z^{-1}[(zI - \Phi)^{-1}z \bullet z^{-1}GU(z)]$$

$$= \dots = \sum_{j=0}^{k-1} \Phi^{k-j-1}Gu(j)$$

- Result: $x(t) = \Phi^k x(0) + \sum_{j=0}^{k-1} \Phi^{k-j-1}Gu(j)$

Ex.9-18 Solution:

Derive $(zI-A)^{-1}$:

$$|zI - A| = \begin{vmatrix} z & -1 \\ 0.16 & z+1 \end{vmatrix} = (z+0.2)(z+0.8)$$

$$(zI - A)^{-1} = \frac{\text{adj}(zI - A)}{|zI - A|} = \frac{\begin{bmatrix} z+1 & 1 \\ -0.16 & z \end{bmatrix}}{(z+0.2)(z+0.8)}$$

$$= \frac{1}{3} \begin{bmatrix} \frac{4}{z+0.2} - \frac{1}{z+0.8} & \frac{5}{z+0.2} - \frac{5}{z+0.8} \\ \frac{-0.8}{z+0.2} - \frac{0.8}{z+0.8} & \frac{-1}{z+0.2} + \frac{4}{z+0.8} \end{bmatrix}$$

$$\Phi(k) = A^k = Z^{-1} \left[(zI - A)^{-1} z \right] \text{ discrete State transition matrix}$$

$$= \frac{1}{3} \begin{bmatrix} 4(-0.2)^k - (-0.8)^k & 5(-0.2)^k - 5(-0.8)^k \\ -0.8(-0.2)^k + 0.8(-0.8)^k & -(-0.2)^k + 4(-0.8)^k \end{bmatrix} \quad 40$$

$$u(k) = 1 \quad \text{Z-transform} \quad U(z) = \frac{z}{z-1}$$

$$X(z) = (zI - A)^{-1} [zx(0) + BU(z)]$$

$$= \begin{bmatrix} \frac{(z^2 + 2)z}{(z + 0.2)(z + 0.8)(z - 1)} \\ \frac{(-z^2 + 1.84z)z}{(z + 0.2)(z + 0.8)(z - 1)} \end{bmatrix} = \frac{1}{18} \begin{bmatrix} \frac{-51z}{z + 0.2} + \frac{44z}{z + 0.8} + \frac{25z}{z - 1} \\ \frac{10.2z}{z + 0.2} + \frac{-35.2z}{z + 0.8} + \frac{7z}{z - 1} \end{bmatrix}$$

$$x(k) = Z^{-1} \{X(z)\} = \frac{1}{18} \begin{bmatrix} -51(-0.2)^k + 44(-0.8)^k + 25 \\ 10.2(-0.2)^k - 35.2(-0.8)^k + 7 \end{bmatrix}$$

Assume $k=0,1,2,3$, to double check the results before:

$$x(k) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1.84 \end{bmatrix}, \begin{bmatrix} 2.84 \\ -0.84 \end{bmatrix}, \begin{bmatrix} 0.16 \\ 1.384 \end{bmatrix}$$